

7. In a diffusion experiment, gelatin was mixed with a pink pH indicator and allowed to set. The gelatin was then cut into different sized cubes. The cubes were placed in a beaker containing dilute acid. As the acid diffused through the gelatin the cubes became colourless. The time taken to change from pink to colourless was recorded.

The surface area and volume of the cubes were calculated using the following equations:

$$\text{surface area of cube} = \text{length} \times \text{width} \times \text{number of sides}$$

$$\text{volume of cube} = \text{length} \times \text{width} \times \text{height}$$

- (a) The results of the experiment are shown below.

Length of side of gelatin cube (cm)	Time taken to become colourless (s)	Surface area of cube (cm ²)	Volume of cube (cm ³)	Surface area:volume ratio
1	20	6	1	6:1
5	41	150		6:5
7	104		343	6:7
10	610	600	1000	

- (i) **Complete** the table. [3]
- (ii) Sarah suggested that diffusion of the acid into the gelatin is quicker in the smaller cubes because the time taken for them to become colourless is less. Explain whether you agree with Sarah. [2]

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- (b) Red blood cells transport oxygen around the body to allow respiration to occur in the tissues. Red blood cells have a volume of 90 units and a surface area of 136 units. Explain how the surface area:volume ratio of red blood cells make them suitable for their function. [3]

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THE PERIODIC TABLE

Group

1

2

3

4

5

6

7

0

1 H Hydrogen 1										4 He Helium 2																									
7 Li Lithium 3		9 Be Beryllium 4								11 B Boron 5		12 C Carbon 6		14 N Nitrogen 7		16 O Oxygen 8		19 F Fluorine 9		20 Ne Neon 10															
23 Na Sodium 11		24 Mg Magnesium 12								27 Al Aluminium 13		28 Si Silicon 14		31 P Phosphorus 15		32 S Sulfur 16		35.5 Cl Chlorine 17		40 Ar Argon 18															
39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21		48 Ti Titanium 22		51 V Vanadium 23		52 Cr Chromium 24		55 Mn Manganese 25		56 Fe Iron 26		59 Co Cobalt 27		59 Ni Nickel 28		63.5 Cu Copper 29		65 Zn Zinc 30		70 Ga Gallium 31		73 Ge Germanium 32		75 As Arsenic 33		79 Se Selenium 34		80 Br Bromine 35		84 Kr Krypton 36	
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39		91 Zr Zirconium 40		93 Nb Niobium 41		96 Mo Molybdenum 42		99 Tc Technetium 43		101 Ru Ruthenium 44		103 Rh Rhodium 45		106 Pd Palladium 46		108 Ag Silver 47		112 Cd Cadmium 48		115 In Indium 49		119 Sn Tin 50		122 Sb Antimony 51		128 Te Tellurium 52		127 I Iodine 53		131 Xe Xenon 54	
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57		179 Hf Hafnium 72		181 Ta Tantalum 73		184 W Tungsten 74		186 Re Rhenium 75		190 Os Osmium 76		192 Ir Iridium 77		195 Pt Platinum 78		197 Au Gold 79		201 Hg Mercury 80		204 Tl Thallium 81		207 Pb Lead 82		209 Bi Bismuth 83		210 Po Polonium 84		210 At Astatine 85		222 Rn Radon 86	
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89																															
																	Key																		

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number